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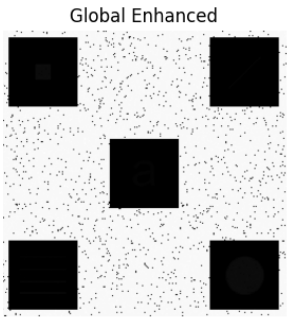
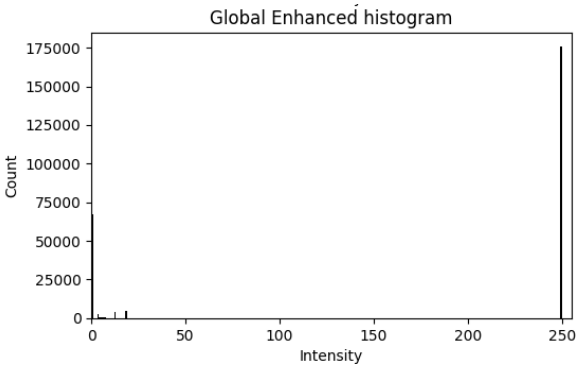
1. Please use a Histogram Statistics method and a local enhancement method to extract the hidden image from the image 'hidden object.jpg.' Please describe your parameters and method in detail, and print out the source code.

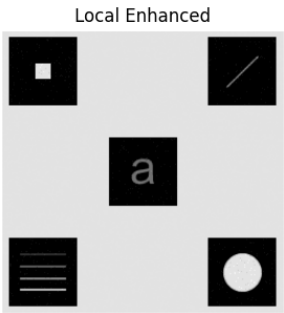
<u>Histogram Statistics Method:</u>

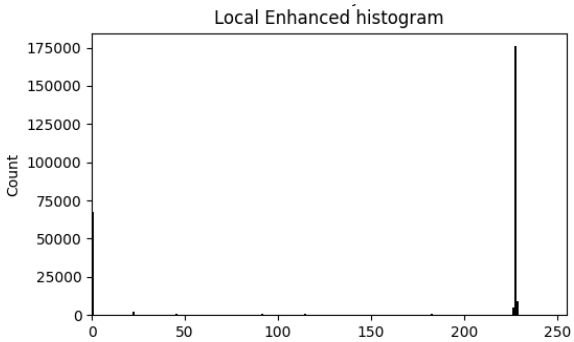
I use the method "Histogram Equalization". What histogram equalization does is to adjust the pixel intensities of the image so that the histogram of the output image is more uniform. Doing so increases the contrast of an image.





<u>Local Enhancement Method:</u>





This method 1st calculates the global mean (μ_{global}) & global standard deviation (σ_{global}) of the whole image. We then multiply the global mean by some k_0 & k_1, & global standard deviation by some k_2 & k_3. Where:

1. k_0 : the lower threshold for the local mean
2. k_1 : the upper threshold for the local mean
3. k_2 : the lower threshold for the local standard deviation
4. k_3 : the upper threshold for the local standard deviation

A local kernel of size n*n slides across the image for each pixel. When the kernel is at a pixel, the local mean (μ_{local}) & local standard deviation (σ_{local}) of the pixels in the n*n kernel are calculated. And if the local mean & standard deviation satisfy both

$k_0\mu_{global} < \mu_{local} < k_1\mu_{global}, k_2\sigma_{global} < \sigma_{local} < k_3\sigma_{global}$ then we multiply the original pixel intensity of that pixel by a constant C. Mathematically, it looks like this:

$$g(x,y) = \begin{cases} C * f(x,y) \; if \; k_0\mu_{global} < \mu_{local} < k_1\mu_{global} \; \& \; k_2\sigma_{global} < \sigma_{local} < k_3\sigma_{global} \\ f(x,y) \; otherwise \end{cases}$$

I set because $k_0 = 0.0, k_1 = 0.1, k_2 = 0.0, k_3 = 0.1$ $because$ they focus enhancement on very dark, low-contrast regions where hidden details are likely buried, while leaving bright areas unchanged. The low values ensure only the darkest, flattest areas are boosted, and $C = 22.8$ gives enough amplification to reveal details, while not over-amplifying the noise, as when C gets larger, the hidden objects do get brighter, but the noise is also significantly more visible.

2. Please comment and compare the Histogram Statistics method and the local enhancement method. (20)

The Histogram Statistics method adjusts the entire image based on its overall intensity distribution, using global mean and variance to improve brightness and contrast uniformly. It's simple and effective for correcting general under- or over-exposure, but cannot adapt to local variations, so dark regions may remain hidden while bright areas can get over-enhanced.

In contrast, the local enhancement method computes mean and variance within small neighbourhoods, allowing it to adjust contrast adaptively for each region. This makes it better at revealing details in locally dark or low-contrast areas while preserving bright regions.